

KURGAN, Russian Federation

WMO: 286610

Lat: 55.4562N

Lon: 65.4012E

Elev: 237

StdP: 14.57

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.7	-21.1	-32.1	0.9	-26.5	-26.6	1.3	-20.8	23.3	18.4	21.2	16.4	5.0	180	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	88.7	66.8	85.2	66.1	82.0	64.9	70.1	83.2	68.6	80.9	66.8	78.1	9.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	96.1	74.6	64.2	90.8	73.3	62.6	85.7	71.9	34.2	83.4	32.9	80.8	31.5	77.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.8	20.0	17.9	-31.6	93.3	6.5	3.3	-36.2	95.6	-40.1	97.6	-43.7	99.5	-48.4	101.9		
			-31.7	72.8	6.5	2.4	-36.4	74.5	-40.2	76.0	-43.8	77.3	-48.5	79.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	2.9	6.3	21.7	41.1	55.5	64.3	66.8	63.4	52.4	39.2	21.7	8.7
	DBStd	25.13	12.72	12.74	9.92	10.26	9.52	8.24	7.01	7.50	8.46	8.67	13.19	13.56
	HDD50	6465	1460	1222	877	300	54	5	0	2	69	348	848	1280
	HDD65	10470	1925	1642	1342	717	318	110	65	123	386	799	1298	1745
	CDD50	1785	0	0	0	34	225	434	520	416	142	14	0	0
	CDD65	316	0	0	0	1	24	89	120	73	9	0	0	0
	CDH74	3502	0	0	0	16	359	1040	1259	707	121	0	0	0
	CDH80	1146	0	0	0	2	96	365	421	232	30	0	0	0
	WSAvg	8.3	7.6	8.1	9.0	9.4	9.8	8.4	7.7	7.6	8.1	8.1	8.3	7.9
	PrecAvg	15.0	0.8	0.5	0.5	0.8	1.3	2.1	2.3	2.2	1.4	1.2	1.0	0.8
(15) Precipitation	PrecMax	19.7	1.7	1.2	1.6	2.2	3.3	5.2	4.6	4.5	3.5	4.3	1.8	1.9
	PrecMin	8.7	0.0	0.0	0.0	0.0	0.2	0.2	0.1	0.2	0.4	0.2	0.0	0.0
	PrecStd	2.8	0.4	0.3	0.4	0.6	0.7	1.2	1.1	1.1	0.7	0.9	0.5	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	36.8	48.4	75.5	87.4	92.1	92.9	91.5	84.0	67.7	48.1	36.5
		MCWB	32.0	33.6	41.9	55.8	62.8	67.8	68.2	68.3	62.1	53.0	43.1	34.2
	2%	DB	29.6	32.6	40.8	68.5	82.5	88.2	88.2	86.2	77.0	61.6	43.2	33.5
		MCWB	28.4	30.8	36.1	51.8	60.7	66.2	67.6	67.0	59.3	50.1	40.2	31.8
	5%	DB	25.8	28.7	37.5	63.0	77.8	84.3	84.6	81.6	71.1	56.7	39.6	30.3
		MCWB	24.6	27.1	34.0	49.4	58.8	65.8	67.1	65.4	57.7	47.9	37.1	28.9
	10%	DB	21.5	24.6	35.1	57.5	73.2	80.1	81.2	77.3	66.3	52.3	36.6	26.7
		MCWB	20.5	23.1	32.4	47.4	57.0	63.9	66.3	63.9	55.5	46.0	34.6	25.5
	0.4%	WB	32.3	33.8	41.3	56.8	66.6	71.7	72.0	71.1	64.8	54.9	44.2	34.4
		MCDB	33.1	35.8	47.2	70.8	83.7	86.5	83.3	83.4	79.8	63.9	46.6	35.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.5	31.0	36.7	53.4	62.5	69.2	70.4	69.0	60.9	51.7	40.3	32.1
		MCDB	29.6	32.8	39.9	65.6	76.3	82.7	82.8	81.8	72.6	59.5	42.6	33.6
	5%	WB	24.7	27.2	34.6	50.7	60.5	67.1	69.1	67.0	58.7	48.7	37.4	29.0
		MCDB	26.0	28.6	37.1	61.9	74.8	79.9	81.3	77.9	69.1	55.0	39.5	30.2
	10%	WB	20.5	23.2	32.7	47.7	58.1	65.0	67.4	65.1	56.4	46.4	34.7	25.5
		MCDB	21.5	24.6	35.0	57.2	70.8	77.2	78.4	74.7	64.8	52.1	36.4	26.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	15.7	15.7	18.2	21.4	20.4	19.7	19.0	17.6	13.6	11.3	12.1
		MCDBR	14.1	14.7	14.4	26.9	27.9	26.5	25.3	25.7	25.4	22.4	11.8	12.0
		MCWBR	13.7	13.7	10.8	13.5	11.8	10.6	9.0	10.0	11.5	12.9	9.6	11.2
		MCDBR	14.2	14.6	13.5	24.5	25.4	24.1	21.9	22.3	22.7	20.0	11.9	11.7
	5% WB	MCWBR	14.1	13.7	10.5	13.1	11.8	10.5	9.4	9.9	11.6	12.6	10.2	11.1
(40) Clear-Sky Solar Irradiance	taub		0.289	0.320	0.376	0.416	0.416	0.410	0.429	0.443	0.390	0.360	0.320	0.289
	taud		2.048	2.017	1.944	2.101	2.206	2.297	2.245	2.172	2.279	2.231	2.132	2.019
	Ebn at Noon		187	227	241	253	262	265	257	243	240	214	174	153
	Edh at Noon		21	32	44	44	42	39	40	41	32	26	20	17
(44) All-Sky Solar Radiation	RadAvg		282	617	1109	1417	1727	1872	1773	1398	945	478	240	181
	RadStd		19	30	63	152	150	166	146	137	79	79	31	22

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page